Calder Flow Systems | Product engineering & plant file

ARCHIVED COPY | FICTIONAL CASE RECORD | References and identities are simulated.

Specification register | CF-RP 380 series

Product family	Single-stage radial-flow centrifugal pump; one suction inlet; discharge outlet 40 mm; single-entry impeller.
Revision register	MK-I / MK-II / MK-III; internal BOM and workshop orders carry separate component revision codes.
Engineering custodian	Calder Flow Systems, Manufacturing Engineering / Manchester

A product-series number denotes sales envelope and output specification. Engineering change notices record imported core configurations separately from the finished pump number. Changes were introduced during the life of each sales revision; do not assume that a product sales revision uniquely identifies the core received in UK stores.

Core drawings | engineering descriptions

Archive	As-delivered mechanical configuration	UK activities required
SA-4 / drawing PV-C104	Machined volute housing without fitted impeller, shaft bearing pack or rotor. Unsealed cavity; wet-circuit test not practicable before UK fitting.	Install impeller, shaft, bearings and seals; align motor; machine mounting faces; pressure and flow test.
PB-7 / drawings PV-C107 and PV-C207	Volute, impeller, shaft, bearings and mechanical seal carried on an aligned wet end. Chinese vendor test uses an external drive fixture. No plant motor, control unit or final base.	Fit drive motor and electrical interface, machine base, assemble cooling loop, integrate instrumentation and full load test.
PB-8 / drawings PV-C208 and PV-C308	Pre-aligned wet end. Impeller and sealed shaft module retained; hydraulic acceptance test performed at vendor on externally driven stand.	Machine adapter and drive flange; fit motor/controller and safety systems; balance, calibrate and full-load test.
SA-5 / drawing PV-C305	Volute shell and loose impeller; no assembled shaft or bearing housing; rotating assembly fitted at Manchester.	Complete wet end; machine bearing saddle; align rotor; motor and control installation; performance verification.
PB-9 / drawing PV-C409	Hydraulically assembled core with rotor, bearings, mechanical seal, inlet and discharge ports; vendor rig report records measurable flow under external drive.	Motor, electrical power, control, support frame, final drive-coupling integration; pressure and endurance testing.
SA-6 / drawing PV-C406	Unassembled stator shell and loose wear parts; lacks impeller, bearing assembly and functioning rotor.	UK installation and balancing of impeller/rotor; seal integration, castings adjustment and final testing.

Revision archive | internal correspondence

6 May 2025 - manufacturing engineer	The procurement department continues to use the vendor item heading “pump cassette” and code 8413.91 on all order confirmations. The workshop drawings distinguish completed wet ends from housings awaiting their rotating assemblies.
11 February 2026 - engineering lead	For the 2026 output, drawings PV-C409 and PV-C406 are both released against MK-III commercial pump contracts; the installation times and wet-end test stages are not comparable.

The above technical descriptions are factory records, not a decision issued under a tariff-classification ruling.

Calder UK operations | routing extracts

Routing group	Work performed	Plant evidence
P-8 / PB variants	CNC machining of adapter faces; fitting of drive motor, electrical controls and couplings; pressure testing; flow-rate tuning; full-line performance documentation.	CNC cell log B/14; dynamic tests W-71; assembly bay 4 records; technician authorisations.
S-4 / SA variants	CNC impeller and housing finishing; bearing and shaft installation, complete rotor alignment; electrical and hydraulic assembly; endurance tests.	Machining route F/20; pump stand tests W-72; shift sign-off records.

Lot-to-routing excerpt

Shop order group	Affected works orders	Routine observation
P-8	P-2302-19, P-2311-06, P-2311-12, P-2408-12, P-2408-17, P-2503-18, P-2509-09, P-2509-14, P-2602-03	Variable imported core revision, consistent UK P-8 routing group.
S-4	P-2302-14, P-2503-22, P-2602-08	UK adds hydraulic rotating assembly before full pump release.